

High-resolution wave modelling in the Mediterranean Sea: Sensitivity to surface current forcing



Stefanos Ziovas*, John Karagiorgos, Sarantis Sofianos, Vassilios Vervatis

Department of Physics, National and Kapodistrian University of Athens, Greece

Ocean Physics And Modelling Group [<http://www.oc.phys.uoa.gr>]

[stziovas@phys.uoa.gr]



1. Objectives

- Assessment of Mediterranean wave dynamics using a high-resolution (1/36°) WAVEWATCH III model.
- Role of ocean surface currents in modulating wave evolution and associated dominant mechanisms.

2. Data & methodology

Twin numerical simulations to isolate the ocean surface current effect.

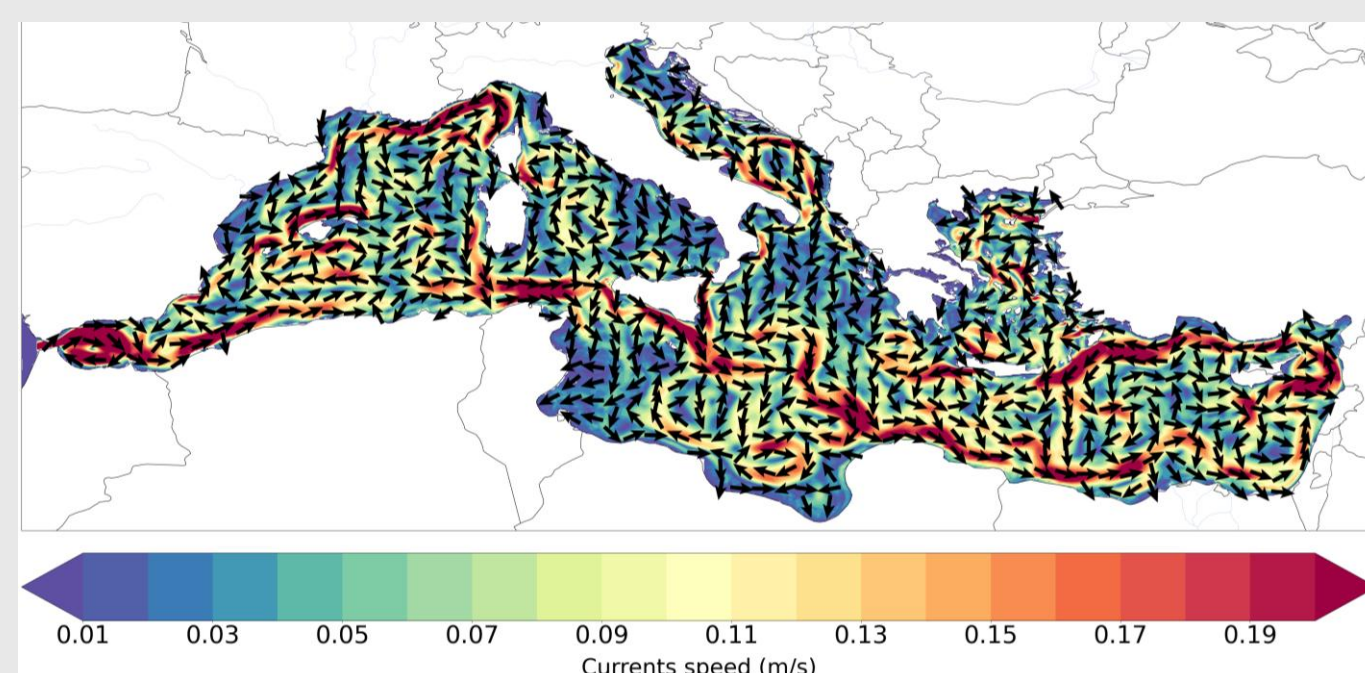


Fig. 1: Study domain and annual currents (m/s).

Model	WAVEWATCH III v6.07 [1]
Horizontal resolution	$\Delta x = \Delta y \approx 1/36^\circ (\approx 3 \text{ km} \times 3 \text{ km})$
Time step	Global $\Delta t = 360 \text{ s}$
Spectral discretization	32 frequencies ($f_0 = 0.04118 \text{ Hz}$, inc. = 1.1)
Directional resolution	24 directions ($\Delta\theta = 15^\circ$)
Physics schemes	PR3, UQ, ST4, DIA, LN1, BT1, DB1, REF1 [2]
Wind forcing	ERA5 reanalysis, 1 h (U_{10} , V_{10})
Surface currents	CMEMS Mediterranean reanalysis, 1 h (u , v)

3. Comparison against observations

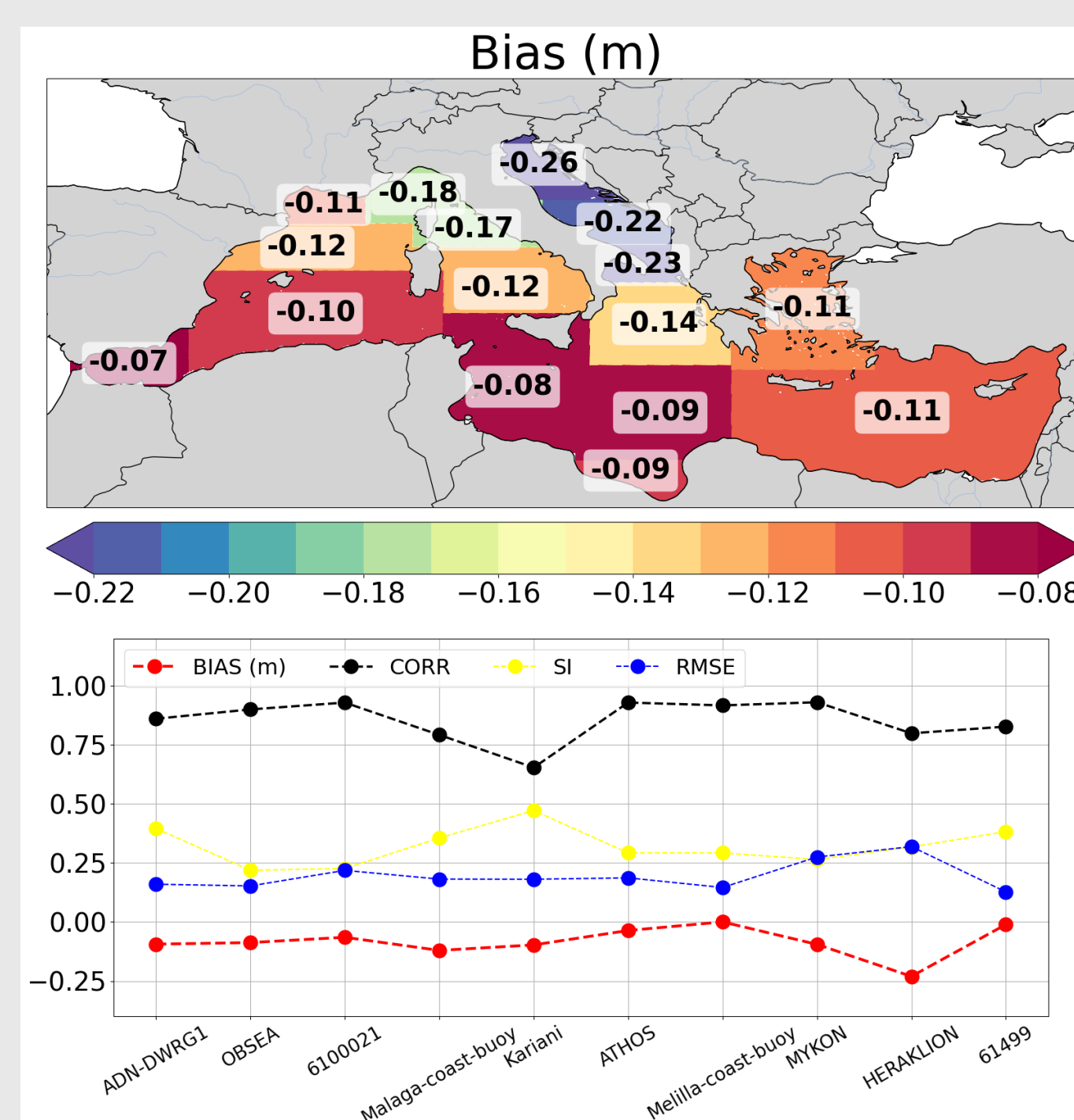


Fig. 2: Comparison of control run (no-current) against altimeter (top) and buoy (bottom) obs.

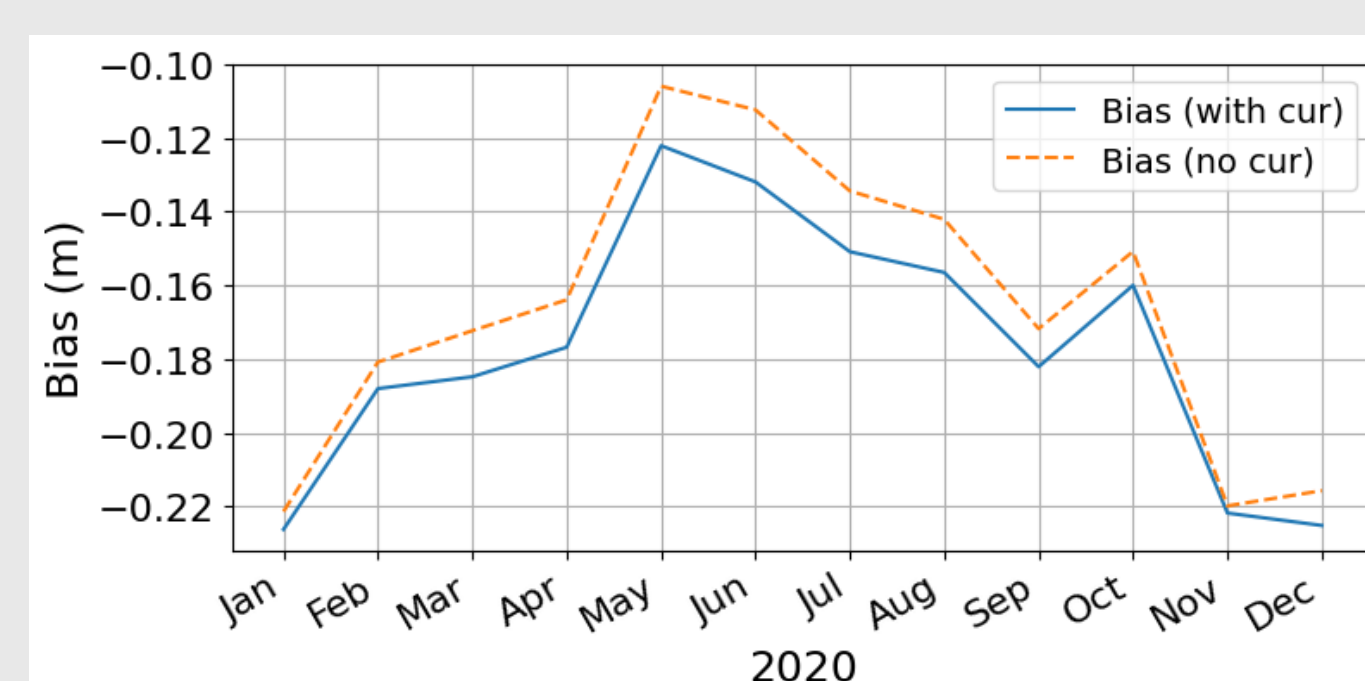
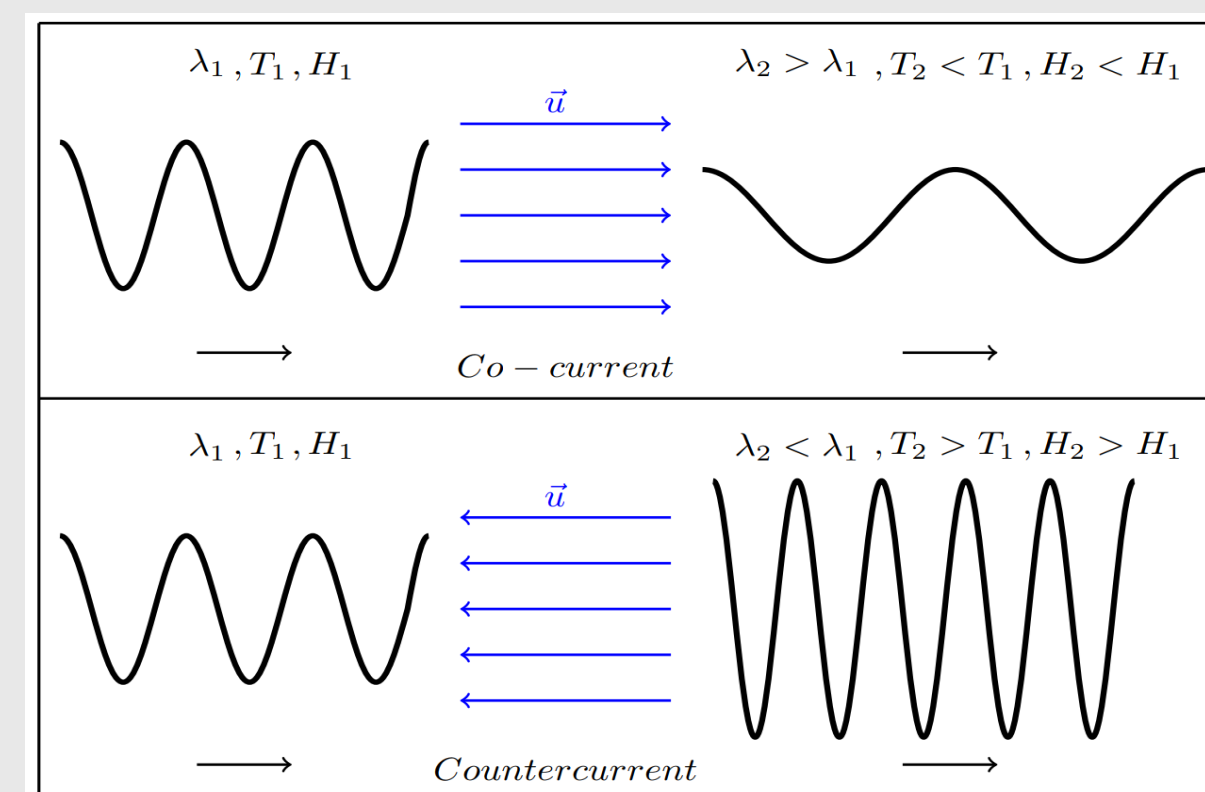


Fig. 3: Comparison against altimeter observations.

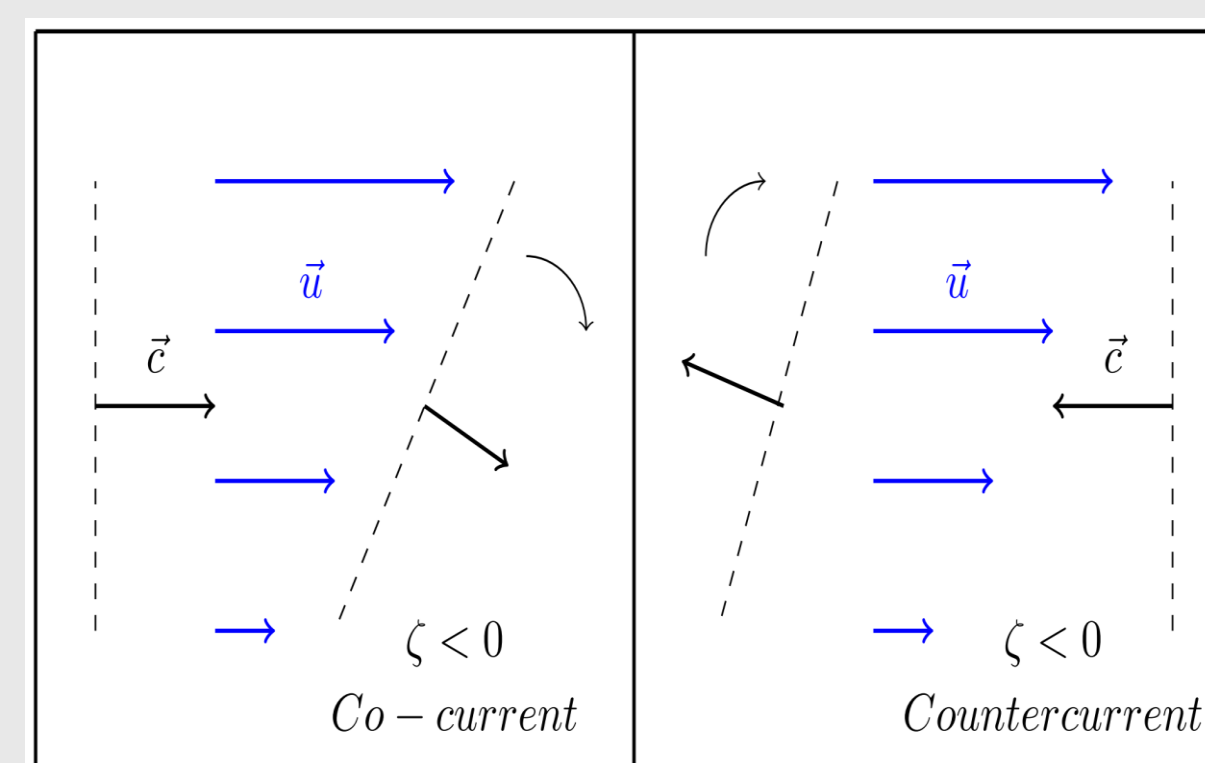
- Inclusion of currents: Moderate impact on model performance.
- Weakest agreement in the Adriatic Sea, associated with short-fetch limitations [3].

4. Theory

Advection



Refraction



5. Results

A. Basin scale

- Current forcing: Reduced SWH, mainly coastal exceptions.
- Refraction: Dominant mechanism (Reg. A).
- Advection & relative wind: Relevant in strong-current, comparable divergence-vorticity (Reg. C,D,E) and sufficient fetch (Reg. B).

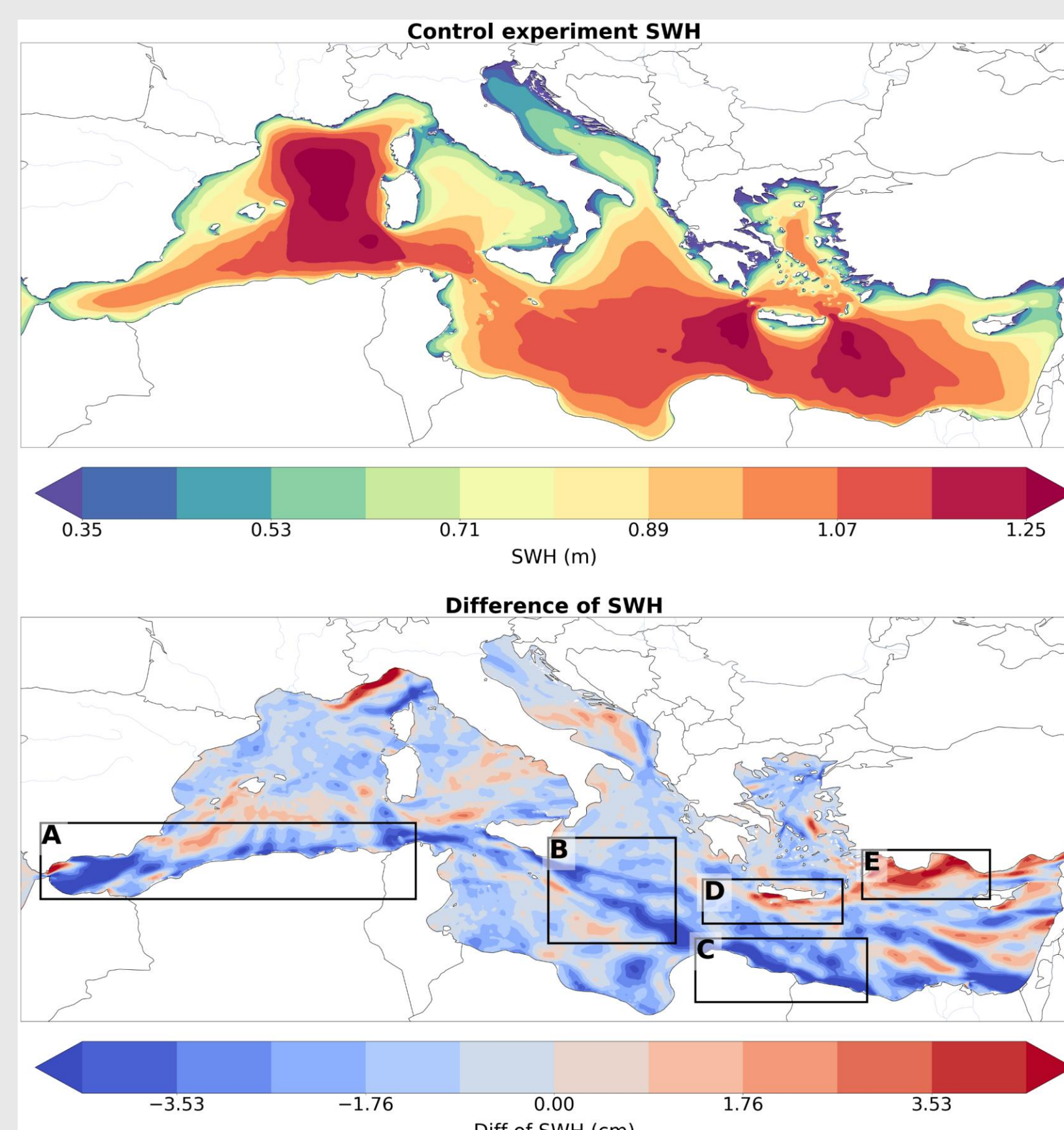
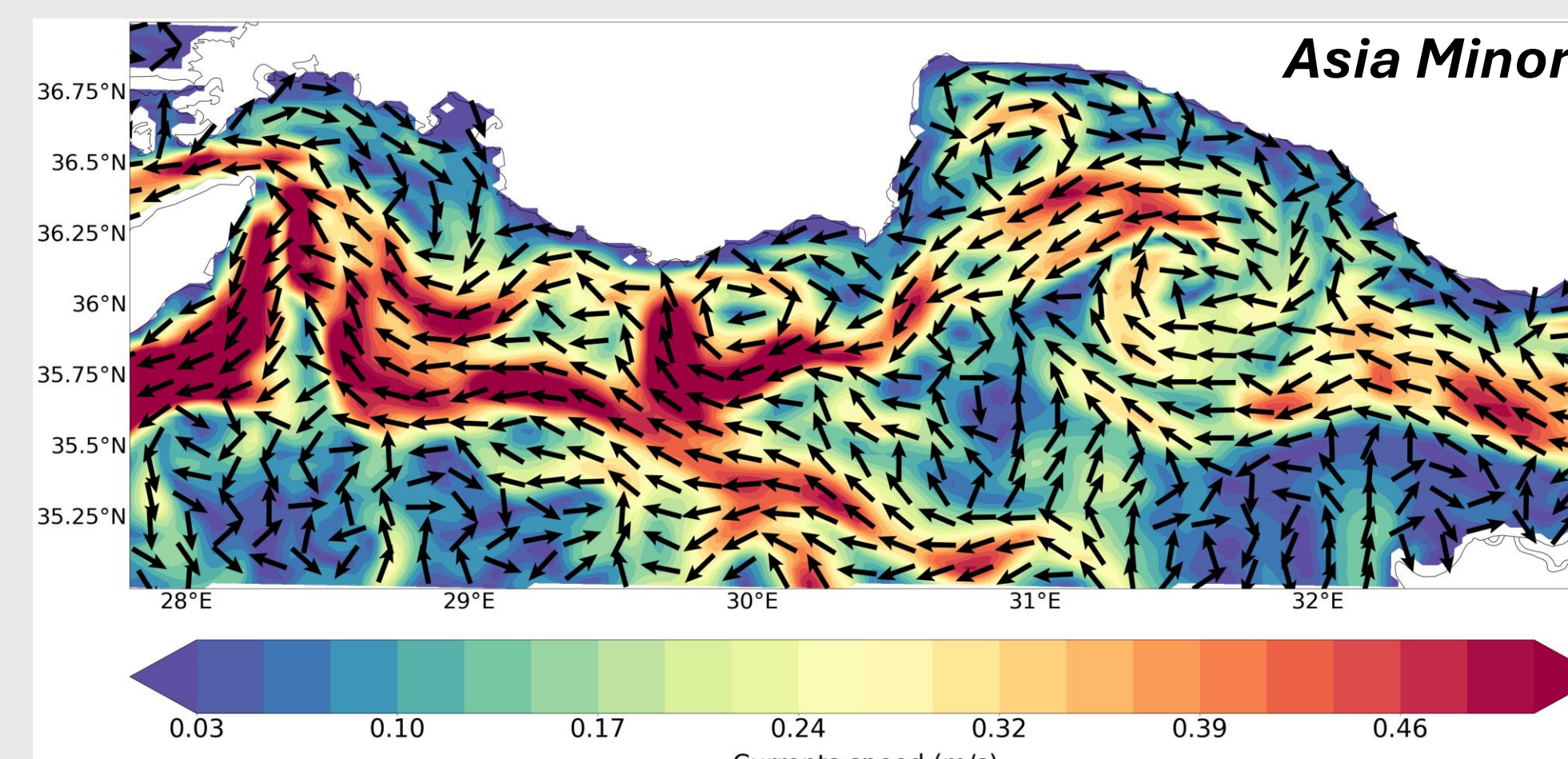


Fig. 4: SWH (control run; top); ΔSWH (bottom).

B. Local scale



- Advection: Opposing currents enhance SWH; co-currents reduce it (Reg. A-C).
- Refraction: Wave-action convergence/divergence causing local SWH amplification or attenuation (Reg. D).
- Relative wind: Co-currents weaken wind input, reducing SWH (Fig. 6).

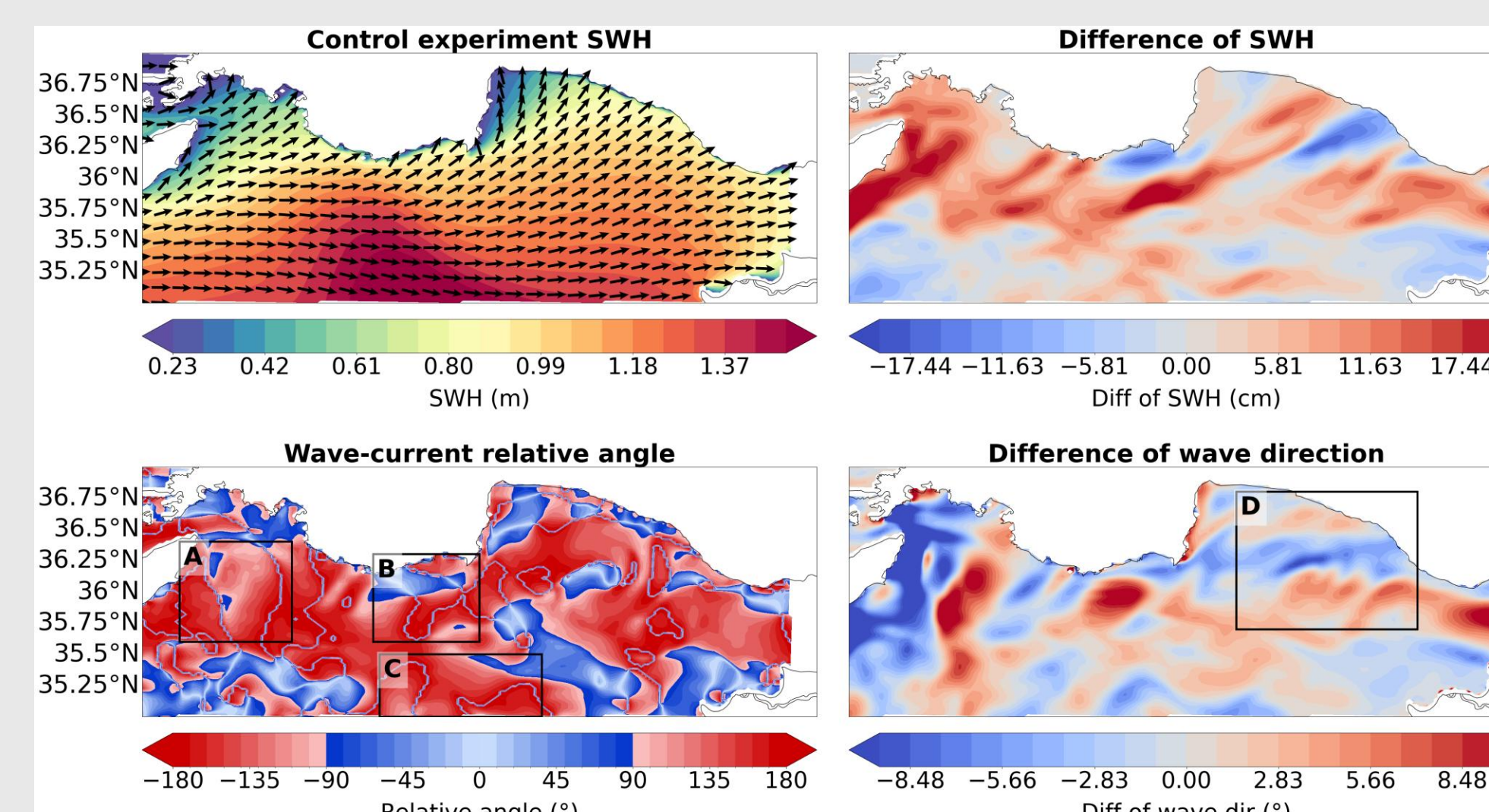


Fig. 5: SWH (control experiment, top left); ΔSWH (top right); wave-current relative angle (bottom left); change in wave direction (bottom right).

Wave direction

- Co-currents: Wave stretching; counter-currents: Wave steepening.
- Shear-induced deformation: Wave refraction driven by current vorticity.

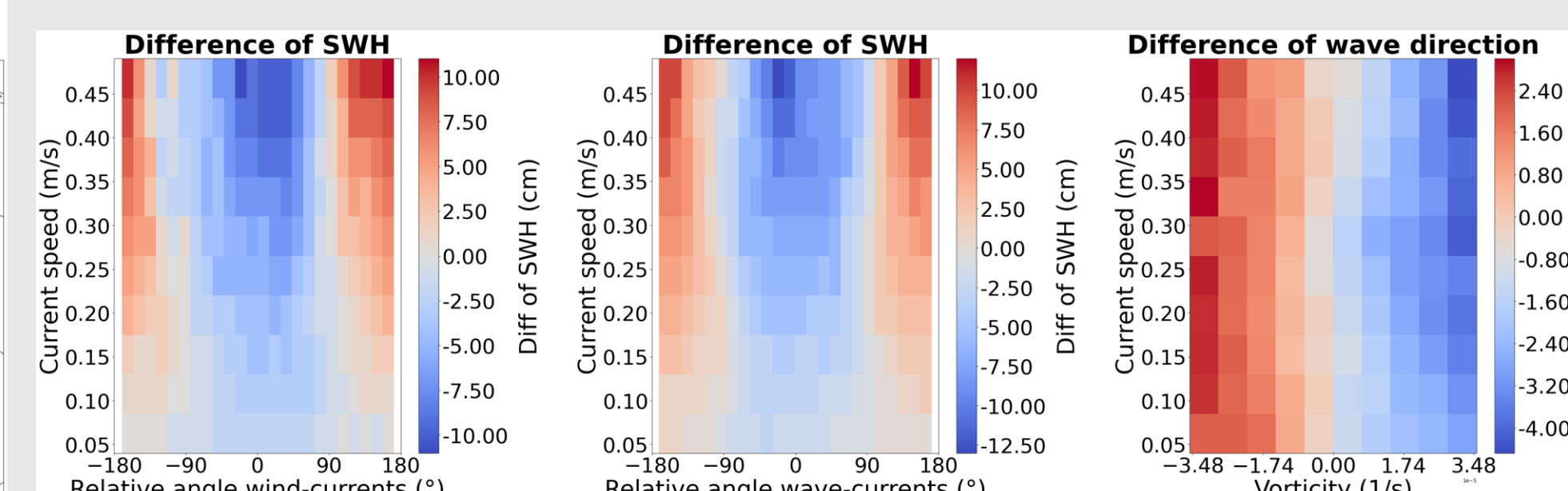


Fig. 6: Mean ΔSWH and wave-direction change binned by current speed and relative direction. Left: ΔSWH vs wind-current angle; Middle: ΔSWH vs wave-current angle; Right: wave-direction change vs vorticity. Angles of 0° indicate co-flowing conditions, and ±180° opposing flow.

C. Case study: Mediane Ianos

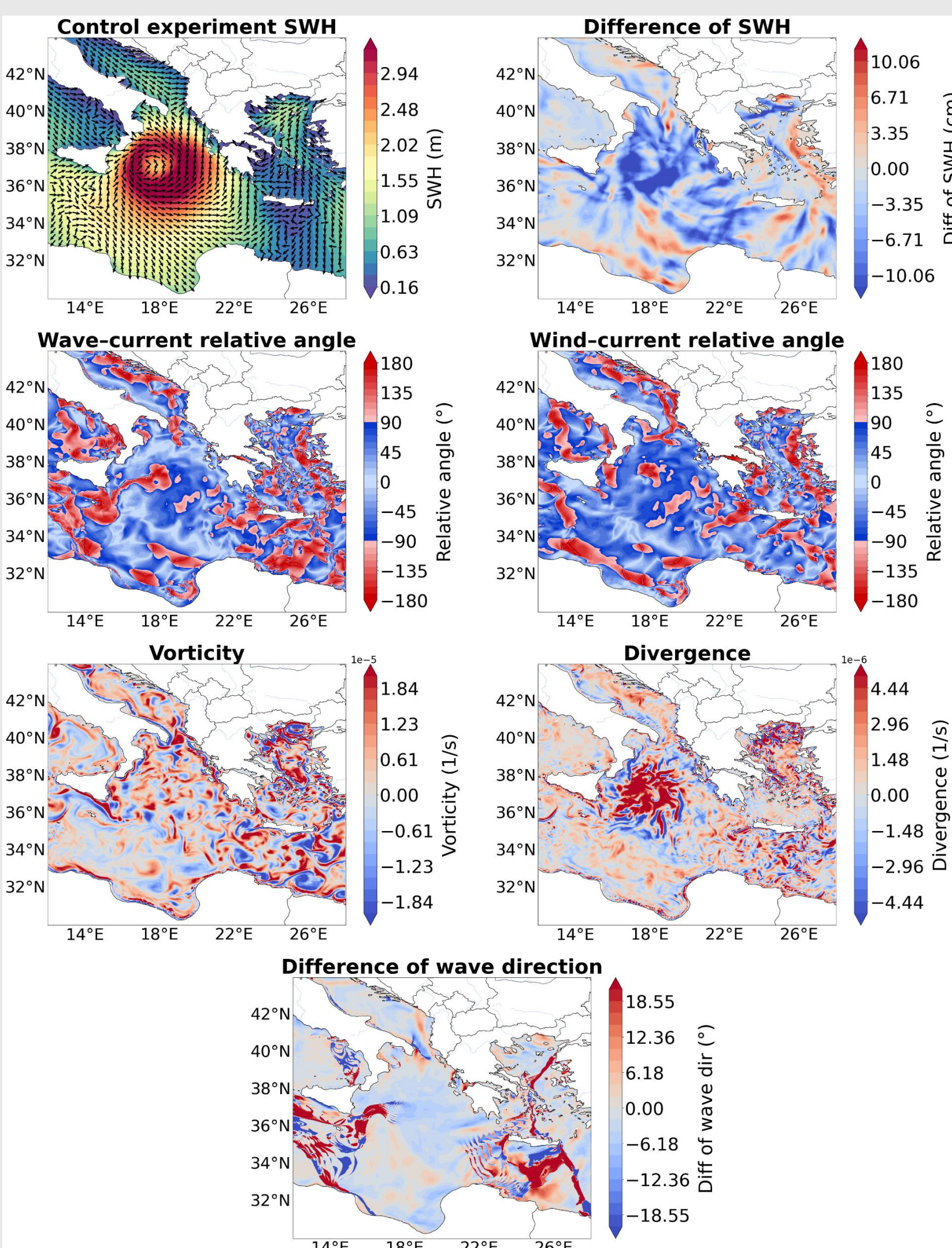


Fig. 7: SWH (control experiment, top left); ΔSWH (top right); wave-current relative angle (2nd row left); wave-current relative angle (2nd row right); current vorticity (3rd row left); current divergence (3rd row right); change in wave direction (bottom).

- Refraction: Weak due to outward-diverging currents near the cyclone center.
- Advection & relative wind: Dominant mechanisms.
- Current-wave-wind alignment: Net SWH reduction.

6. Conclusions & future work

- Ocean currents modulate surface waves by altering propagation and energy balance. Refraction dominates mesoscale wave-action redistribution, while advection and relative wind effects control sub-mesoscale wave growth.
- Ongoing activities include an implementation of a two-way coupled wave-ocean framework (NEMO-WW3), allowing the online interaction between wave and ocean model fields.

References

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